

CHALLENGES & SOLUTIONS REPORT

Raritone Summer Internship 2026

Introduction

During the internship, various technologies related to Artificial Intelligence, Machine Learning, Computer Vision, Virtual Try-On Systems, and Fashion Recommendation Systems were explored. While researching these technologies, several challenges were identified that commonly affect the development of intelligent fashion applications.

This report summarizes the major challenges observed and the possible solutions that can improve system performance and user experience.

1. Challenge: Accurate Body Detection

Description

Human body detection serves as the foundation for virtual try-on systems. Variations in image quality, lighting conditions, and user positioning can affect detection accuracy.

Impact

- Incorrect body localization
- Reduced pose estimation accuracy
- Poor virtual try-on results

Solution

- Use high-quality input images.
- Apply image preprocessing techniques.
- Utilize advanced body detection frameworks such as MediaPipe.
- Improve dataset diversity for better model generalization.

2. Challenge: Pose Estimation Variations

Description

Users may appear in different poses, making landmark detection more complex.

Impact

- Incorrect body keypoint detection
- Misaligned garments

- Reduced fitting accuracy

Solution

- Implement robust pose estimation models.
- Train models using diverse pose datasets.
- Improve landmark tracking algorithms.
- Test models under different pose conditions.

3. Challenge: Cloth Segmentation Accuracy

Description

Separating garments from the body and background can be difficult when clothing colors are similar to surrounding objects.

Impact

- Inaccurate segmentation masks
- Poor garment extraction
- Reduced visual quality

Solution

- Use advanced segmentation models such as U2Net and Detectron2.
- Apply image enhancement techniques.
- Utilize detailed human parsing approaches.

4. Challenge: Garment Alignment

Description

Garments must be aligned according to body shape and pose before virtual fitting.

Impact

- Unrealistic clothing placement
- Distorted outputs
- Reduced user confidence

Solution

- Use body landmark information during alignment.
- Implement garment warping techniques.
- Improve geometric matching methods.
- Utilize pose-aware fitting approaches.

5. Challenge: Realistic Virtual Try-On Generation

Description

Creating realistic virtual fitting results requires the integration of multiple AI modules.

Impact

- Unnatural garment appearance
- Visual inconsistencies
- Reduced realism

Solution

- Improve segmentation accuracy.
- Enhance garment warping techniques.
- Use high-resolution virtual try-on architectures.
- Optimize image preprocessing workflows.

6. Challenge: Dataset Availability and Diversity

Description

AI models require large and diverse datasets for effective training and evaluation.

Impact

- Limited model performance
- Reduced adaptability
- Poor generalization

Solution

- Utilize multiple fashion datasets.
- Include different body types and clothing categories.
- Expand dataset diversity.
- Improve data organization and documentation.

7. Challenge: Recommendation Quality

Description

Providing accurate fashion recommendations requires a deep understanding of user preferences and behavior.

Impact

- Irrelevant product suggestions
- Reduced user engagement

- Lower personalization

Solution

- Implement hybrid recommendation systems.
- Analyze user interactions.
- Incorporate fashion trend data.
- Continuously improve recommendation algorithms.

8. Challenge: Real-Time Processing

Description

Many AI-powered fashion technologies require significant computational resources.

Impact

- Increased processing time
- Reduced user experience
- Scalability limitations

Solution

- Use optimized AI models.
- Implement lightweight architectures.
- Utilize cloud-based processing.
- Improve workflow efficiency.

9. Challenge: AR/VR Integration

Description

Developing immersive AR/VR fashion experiences involves complex rendering and tracking systems.

Impact

- Increased development complexity
- Hardware limitations
- Performance issues

Solution

- Utilize modern AR/VR frameworks.
- Optimize rendering pipelines.
- Improve motion tracking systems.
- Conduct extensive testing across devices.

Conclusion

The challenges identified during the internship highlight the complexity of developing intelligent fashion systems. Areas such as body detection, pose estimation, segmentation, garment alignment, recommendation systems, and AR/VR integration require continuous improvement and research.

By adopting advanced AI models, improving datasets, and optimizing workflows, many of these challenges can be effectively addressed. These solutions can contribute to the development of more accurate, realistic, and personalized fashion experiences for users.